222712: quartzite, Lawlers – Mount Ida Road

(Walter Williams Formation, Vetters Subgroup, Kalgoorlie Group, Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Menzies (SH 51-5), Ballard (3039)
MGA Zone 51, 257518E 6788598N

Sampled on 7 April 2018

This sample was collected from an outcrop north of the Mount Ida mining centre, about 3.1 km south of Gregory Well, 1.6 km north of Forrest Belle mine, and 0.2 km east of the Lawlers – Mount Ida Road.

Geological context

The unit sampled is a quartzite interleaved with mafic–ultramafic rocks of the Walter Williams Formation of the Kalgoorlie Group, on the eastern limb of the Kurrajong Anticline (Swager et al., 1990; Rattenbury, 1995; Hill et al., 2001). The Walter Williams Formation consists mostly of peridotite and komatiite, ultramafic schist and komatiitic basalt (Hill et al., 2001). A sample of the Forest Belle Gabbro, collected about 4.4 km to the south, yielded an igneous crystallization age of 2731 ± 8 Ma (GSWA 222720, Wingate et al., 2022). About 32 km to the south, a fine-grained, thinly bedded metasedimentary rock that appears to conformably overlie pyroxene-spinifex-textured basalt at the top of the Walter Williams Formation yielded a maximum depositional age of 2695 ± 5 Ma (GSWA 185901, Wingate et al., 2010).

Petrographic description

The sample is a fine-grained, massive quartzite, consisting of >99% quartz, <1% muscovite, and trace zircon. Parts of the rock are overprinted by very fine-grained zoisite alteration. Quartz is anhedral, mostly equant, transparent, unstrained or weakly strained, and averages 0.15 mm in size. Extremely faint grain boundaries suggest most grains are subrounded or rounded, although compaction and/or metamorphism has produced a moderately well-developed polygonal texture and there is only a tiny amount of interstitial quartz cement. Muscovite forms anhedral to subhedral, moderately well-aligned laths up to 0.15 mm long. Zoisite grains are subhedral, equant to slightly tabular, and up to 0.1 mm in size. The sample is interpreted as a pure quartz arenite, with a closely packed, well-sorted framework that is texturally mature. Alignment of muscovite suggests that the quartzite has been mildly deformed at relatively low grade, although it is unclear how much of the semi-polygonal texture is the product of compaction or induration and how much is due to grain-boundary migration during deformation and metamorphism.

Zircon morphology

Zircons isolated from this sample are mainly colourless and anhedral to euhedral. The crystals are up to 150 µm long, and equant to slightly elongate, with aspect ratios up to 4:1. A few zircons exhibit poorly defined surface pitting and, in cathodoluminescence (CL) images, concentric zoning is ubiquitous and some crystals exhibit sector zoning. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 23–24 October 2020, using SHRIMP-B. Eleven analyses of the M257 standard were obtained during the session, of which 10 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.21% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-one analyses were obtained from 21 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). One analysis indicates high common Pb ($f204$ >1%). The date obtained from this analysis (Group D; Table 1) is unreliable, and is considered not to be geologically significant. The remaining 20 analyses can be divided into three groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group Y comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2695 ± 6 Ma (1σ).

Group S comprises 18 analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2816–2712 Ma.

Group P comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2553 ± 10 Ma (1σ).

Figure 1. Hand specimen image of sample 222712: quartzite, Lawlers – Mount Ida Road

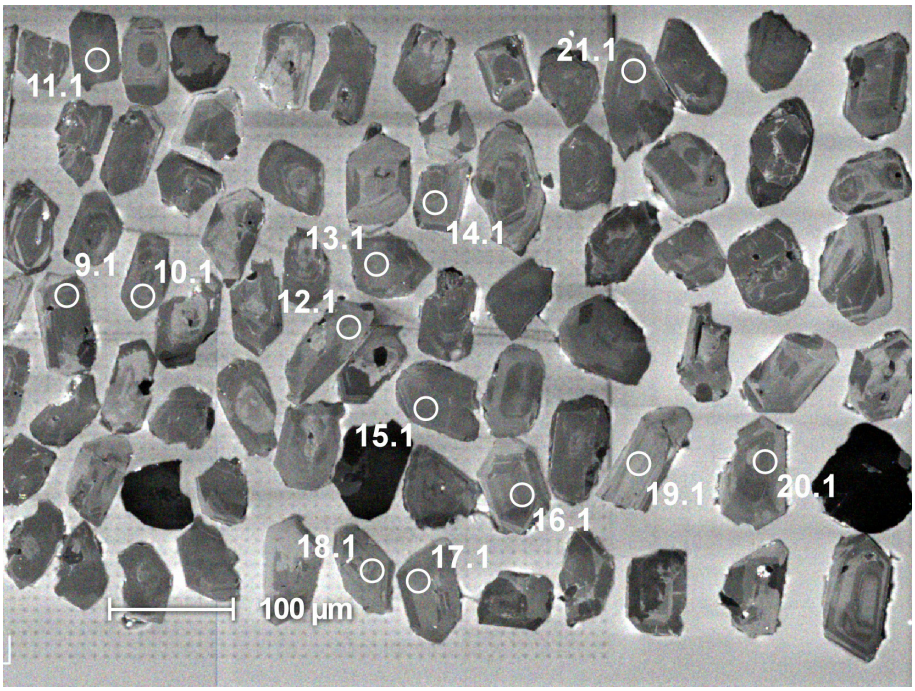

Figure 2. Cathodoluminescence image of representative zircons from sample 222712: quartzite, Lawlers – Mount Ida Road. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 222712: quartzite, Lawlers – Mount Ida Road

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
Y	20	20.1	737	526	0.71	0.198	2.091	0.017	0.18643	0.00065	2.096	0.017	0.18466	0.00072	2515	18	2695	6	6.7
S	16	16.1	594	315	0.53	0.089	2.022	0.017	0.18732	0.00069	2.023	0.017	0.18652	0.00072	2589	18	2712	6	4.5
S	12	12.1	792	506	0.64	0.101	1.954	0.016	0.18760	0.00062	1.956	0.016	0.18669	0.00065	2662	18	2713	6	1.9
S	17	17.1	734	446	0.61	0.068	2.033	0.017	0.18766	0.00062	2.035	0.017	0.18705	0.00064	2577	18	2716	6	5.1
S	9	9.1	647	366	0.57	0.097	1.963	0.017	0.18812	0.00068	1.965	0.017	0.18725	0.00072	2652	19	2718	6	2.4
S	11	11.1	648	461	0.71	0.016	1.982	0.017	0.18850	0.00066	1.982	0.017	0.18836	0.00067	2633	18	2728	6	3.5
S	5	5.1	407	181	0.45	0.057	1.935	0.018	0.18940	0.00082	1.936	0.018	0.18890	0.00084	2685	21	2733	7	1.7
S	13	13.1	1038	776	0.75	0.047	1.921	0.015	0.18941	0.00150	1.922	0.015	0.18899	0.00151	2701	18	2733	13	1.2
S	15	15.1	486	265	0.55	0.014	1.959	0.018	0.18913	0.00075	1.960	0.018	0.18901	0.00076	2658	20	2734	7	2.8
S	21	21.1	679	508	0.75	-0.005	1.922	0.027	0.18903	0.00067	1.922	0.027	0.18907	0.00067	2700	32	2734	6	1.2
S	7	7.1	944	647	0.69	0.014	1.930	0.015	0.18921	0.00055	1.931	0.015	0.18908	0.00055	2691	18	2734	5	1.6
S	6	6.1	466	257	0.55	0.037	1.922	0.017	0.18957	0.00141	1.922	0.017	0.18924	0.00142	2700	20	2736	12	1.3
S	4	4.1	810	528	0.65	0.021	1.937	0.022	0.18968	0.00060	1.937	0.022	0.18949	0.00061	2683	25	2738	5	2.0
S	10	10.1	801	523	0.65	0.010	1.954	0.016	0.18994	0.00064	1.954	0.016	0.18985	0.00064	2665	18	2741	6	2.8
S	14	14.1	782	491	0.63	0.020	1.955	0.016	0.19022	0.00066	1.955	0.016	0.19004	0.00066	2663	19	2743	6	2.9
S	3	3.1	375	177	0.47	0.000	1.998	0.019	0.19005	0.00087	1.998	0.019	0.19005	0.00087	2616	21	2743	8	4.6
S	8	8.1	685	375	0.55	0.026	1.909	0.016	0.19147	0.00066	1.910	0.016	0.19124	0.00067	2715	19	2753	6	1.4
S	1	1.1	803	616	0.77	0.064	1.842	0.031	0.19378	0.00058	1.843	0.031	0.19321	0.00060	2795	39	2770	5	-0.9
S	18	18.1	795	519	0.65	0.020	1.728	0.014	0.19898	0.00061	1.728	0.014	0.19880	0.00062	2943	20	2816	5	-4.5
P	2	2.1	2028	2573	1.27	0.082	2.126	0.016	0.17030	0.00098	2.128	0.016	0.16957	0.00099	2483	16	2553	10	2.7
D	19	19.1	1505	1619	1.08	1.508	2.045	0.015	0.19221	0.00047	2.076	0.016	0.17876	0.00076	2535	16	2641	7	4.0

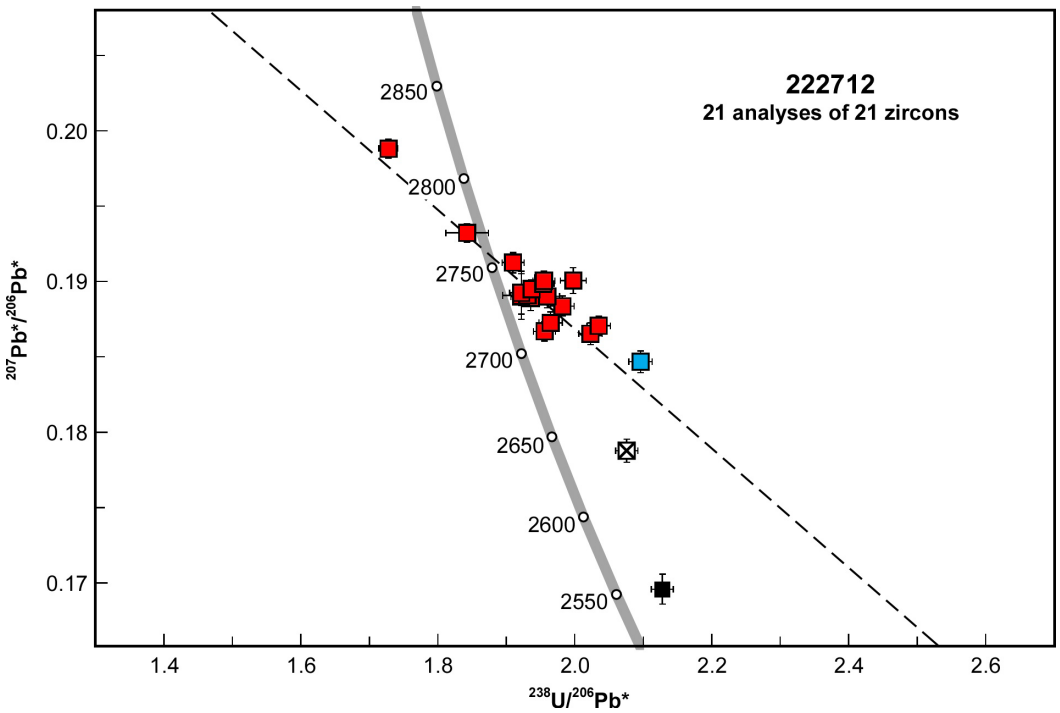

Figure 3. U–Pb analytical data for sample 222712: quartzite, Lawlers – Mount Ida Road. Blue square indicates Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); black square indicates Group P (radiogenic-Pb loss); crossed square indicates Group D (high common Pb)

It is possible that all of the analyses in Groups Y and S are of unmodified detrital zircons, in which case the date of 2695 ± 6 Ma (1σ) for the single analysis in Group Y represents a maximum depositional age. However, the distribution of analyses in Groups Y and S suggests that the zircons have lost variable amounts of radiogenic Pb, and that the $^{207}Pb^*/^{206}Pb^*$ date of 2695 ± 6 Ma (1σ) for the single analysis in Group Y underestimates its age of crystallization.

The combined 19 analyses in Groups Y and S yield a discordia regression with intercepts at 2764 ± 8 and c. 1270 Ma (MSWD = 1.4). The upper intercept of 2764 ± 8 Ma is interpreted as the age of the dominant detrital component, and a conservative maximum age of deposition. The date of 2553 ± 10 Ma (1σ) for the single analysis in Group P is interpreted to reflect loss of radiogenic Pb.

References

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Recommended reference for this publication

Wingate, MTD, Lu, Y, Fielding, IOH and De Paoli, M 2022, 222712: quartzite, Lawlers – Mount Ida Road; Geochronology Record 1855: Geological Survey of Western Australia, 4p.

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